

High-Performance Data Analysis

(Análise de Dados de Elevado Desempenho)

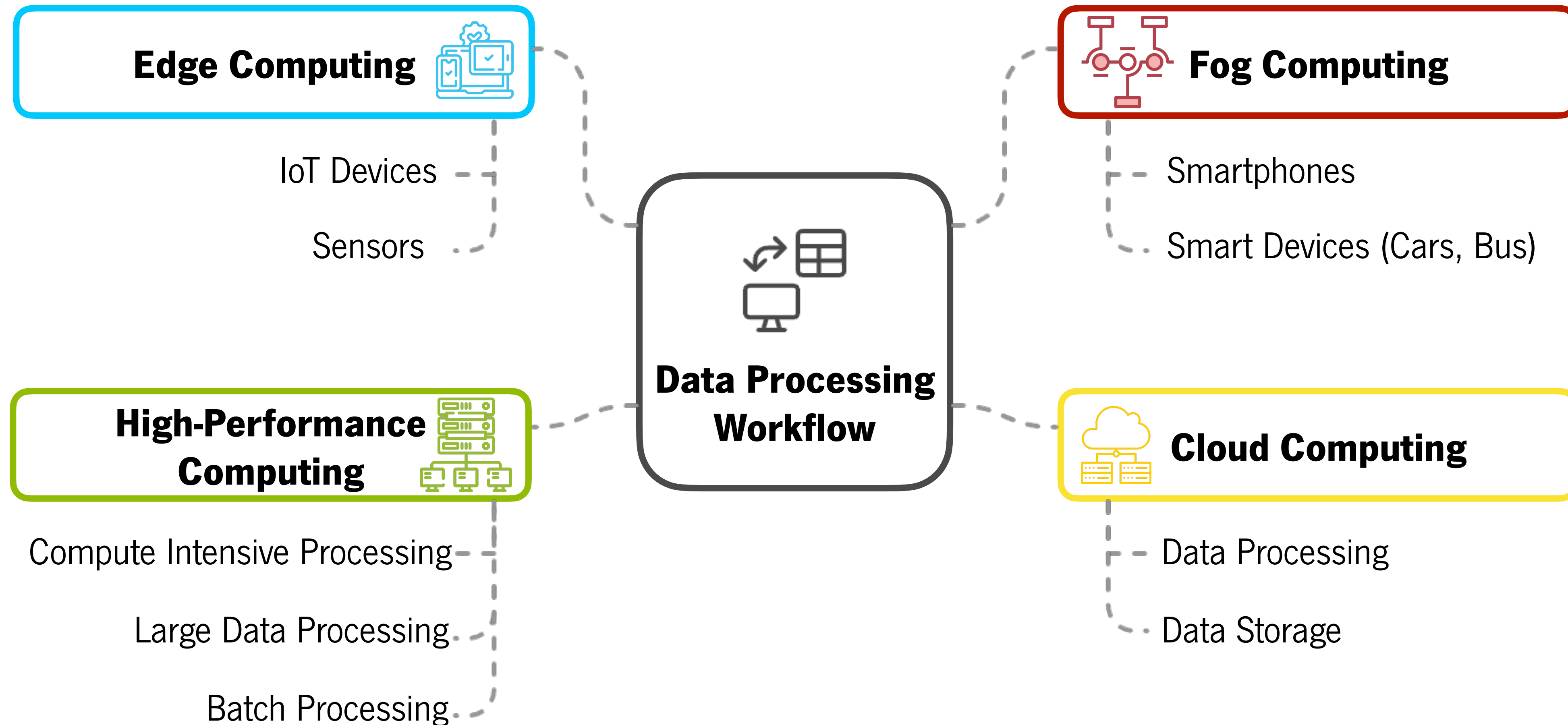
Introduction to Big Data Analysis

University of Minho

2025-2026



Distributed Computing Paradigms



Big Data

Volume

- Refers to the sheer amount of **data generated every second**.
- Large-scale data sources include social media (e.g., X, Facebook), sensors (IoT devices), transactions (financial systems), and multimedia content (videos, images).
- For instance, **Facebook** processes over 4 petabytes of data per day.

Velocity

- Describes the speed at which data is **generated, processed,** and **analyzed**.
- Real-time data streaming requires rapid ingestion and analysis to provide timely insights.
- Technologies such as Apache Kafka, Apache Flink, and Google Dataflow enable fast processing.
- For instance, **stock market** trading systems process millions of transactions per second.

Variety

- **Data comes in multiple formats**: structured (databases, spreadsheets), semi-structured (JSON, XML), and unstructured (videos, images, text).
- Organizations must handle diverse data formats efficiently.
- **Netflix** processes **structured** customer data, **unstructured** movie reviews, and **semi-structured** metadata for recommendations.

Big Data Analysis

❖ What is?

- ▶ The process of examining large and complex datasets to uncover patterns, correlations, trends, and insights.
- ▶ Used to enhance **decision-making, optimize operations, and drive innovation.**

❖ What is the importance?

- ▶ It helps businesses **predict trends, understand customer behavior,** and **improve efficiency.**
- ▶ It is used in various industries like healthcare, finance, retail, and entertainment.

❖ What methods?

- ▶ **Descriptive Analytics** → What happened? (Summarizing past data).
- ▶ **Predictive Analytics** → What might happen? (Using AI & ML models).
- ▶ **Prescriptive Analytics** → What should be done? (Recommending actions).

Big Data Analysis

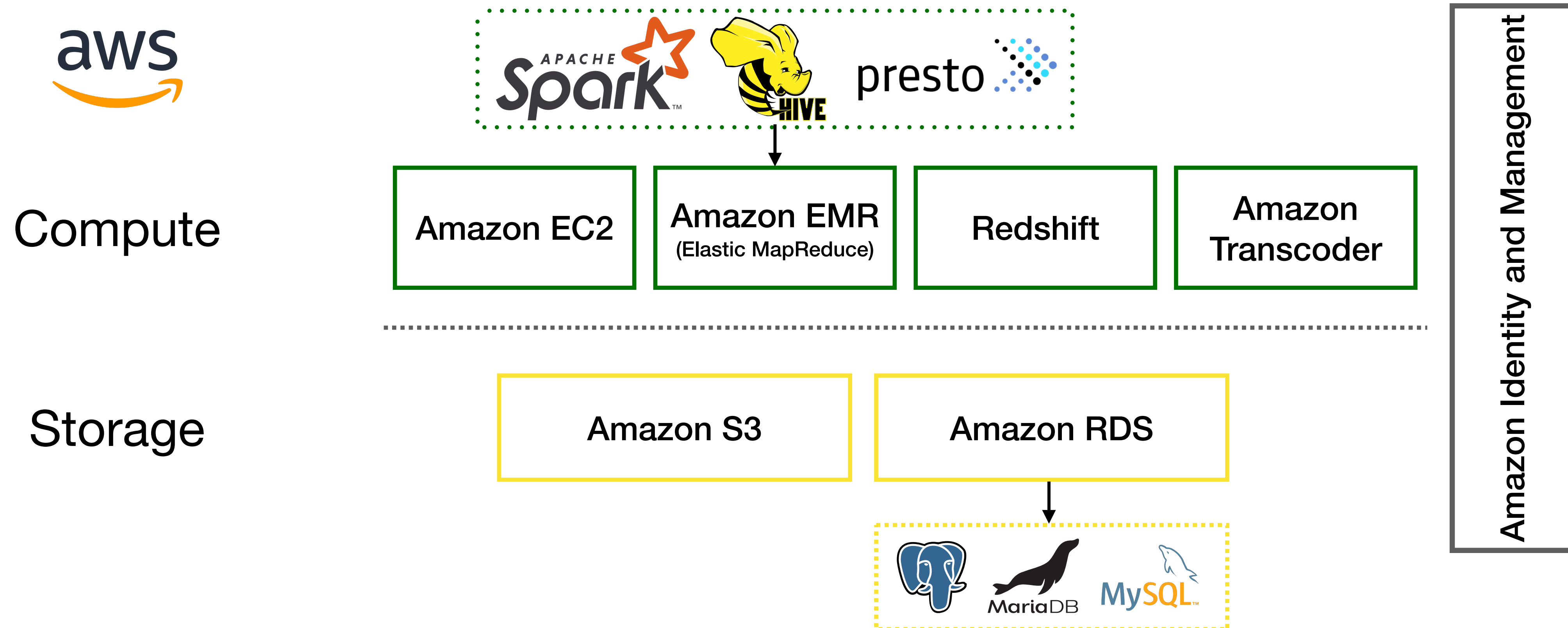
❖ Key Technologies:

- **Data Processing:** Apache Spark, Hadoop, Google Dataflow
- **Storage Solutions:** AWS S3, Google BigQuery, Hadoop HDFS
- **Analytics & Querying:** Presto, Redshift, Looker, Tableau
- **Machine Learning Frameworks:** TensorFlow, Scikit-Learn, Vertex AI, PyTorch

Big Data Analysis - Use Cases



❖ Use Case 1: **Rovio** on Cloud Storage and Infrastructure

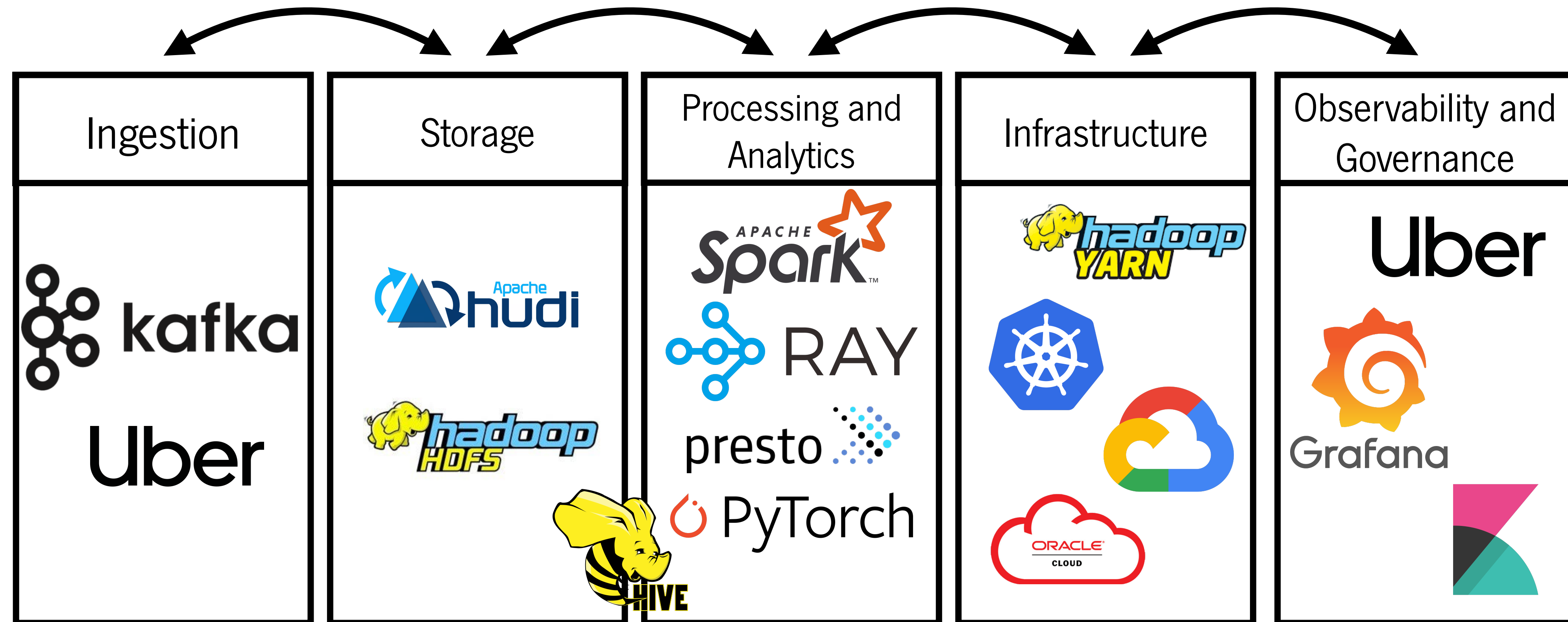


<https://aws.amazon.com/solutions/case-studies/rovio-gaming-service-platform/>

Big Data Analysis - Pipelines

Uber

❖ Use Case 2: **Uber**'s Data and ML Platform



<https://www.uber.com/en-HK/blog/modernizing-ubers-data-infrastructure-with-gcp/>

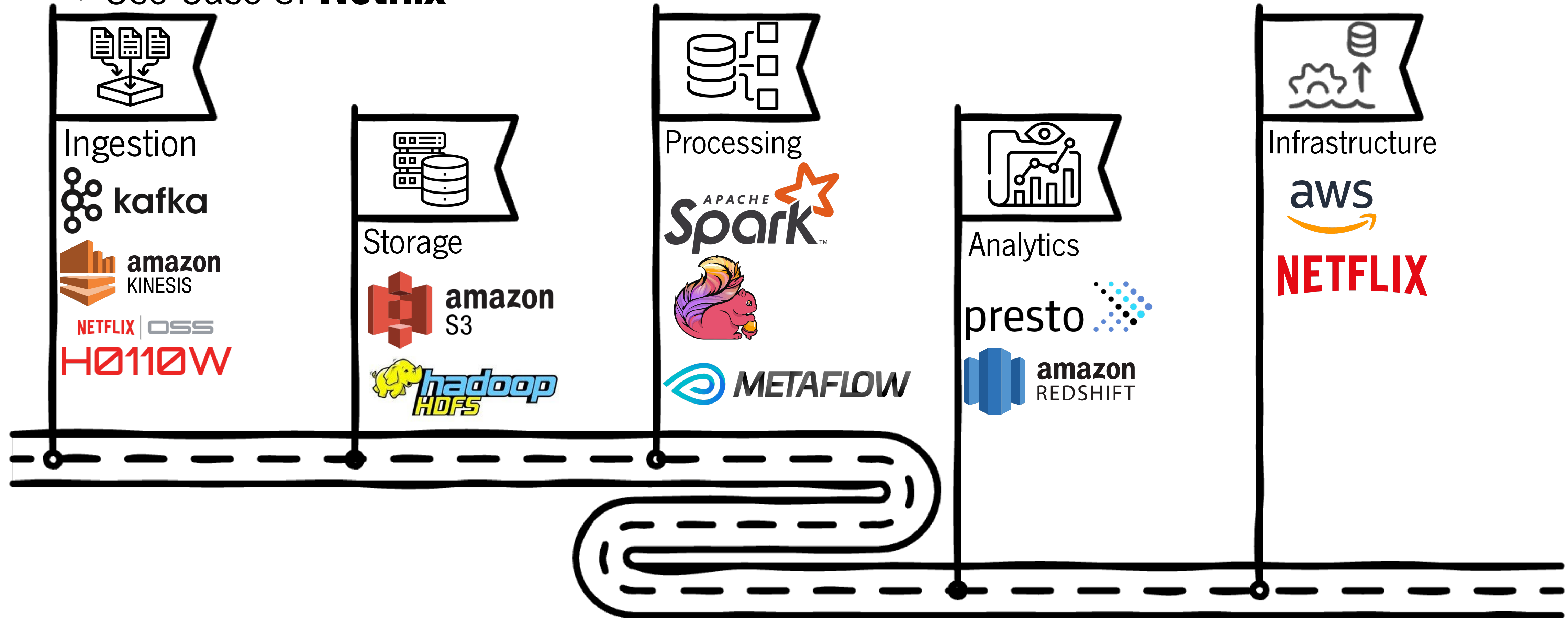
<https://www.uber.com/en-HK/blog/from-predictive-to-generative-ai/>

<https://www.uber.com/en-HK/blog/how-uber-uses-ray-to-optimize-the-rides-business>

Big Data Analysis - Pipelines

NETFLIX

❖ Use Case 3: Netflix



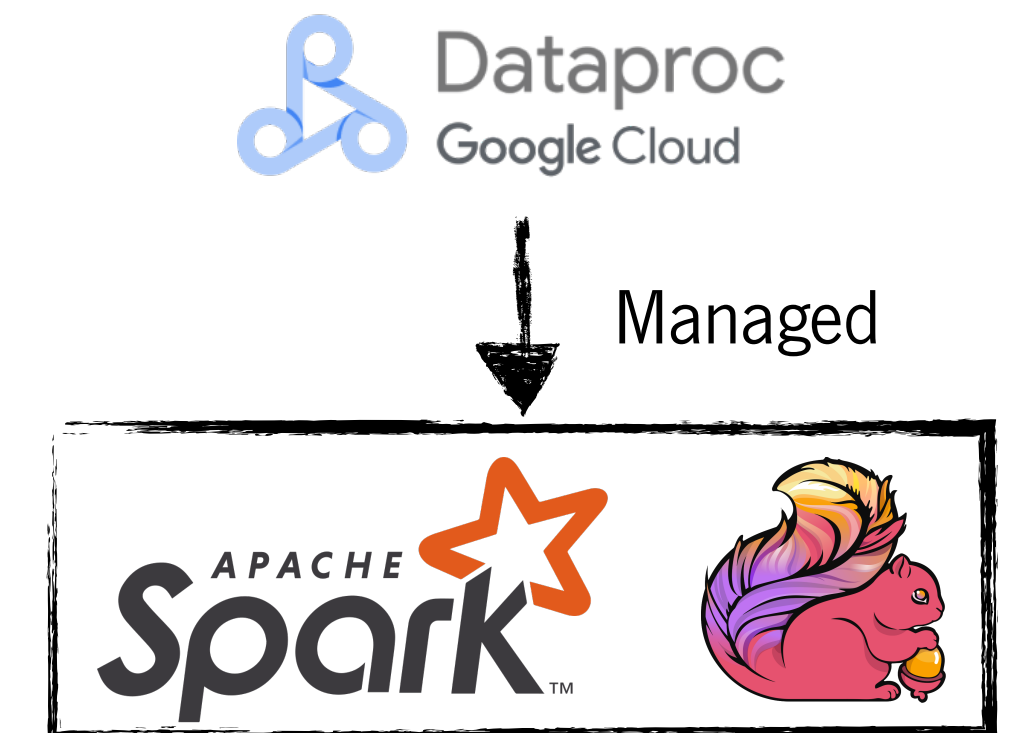
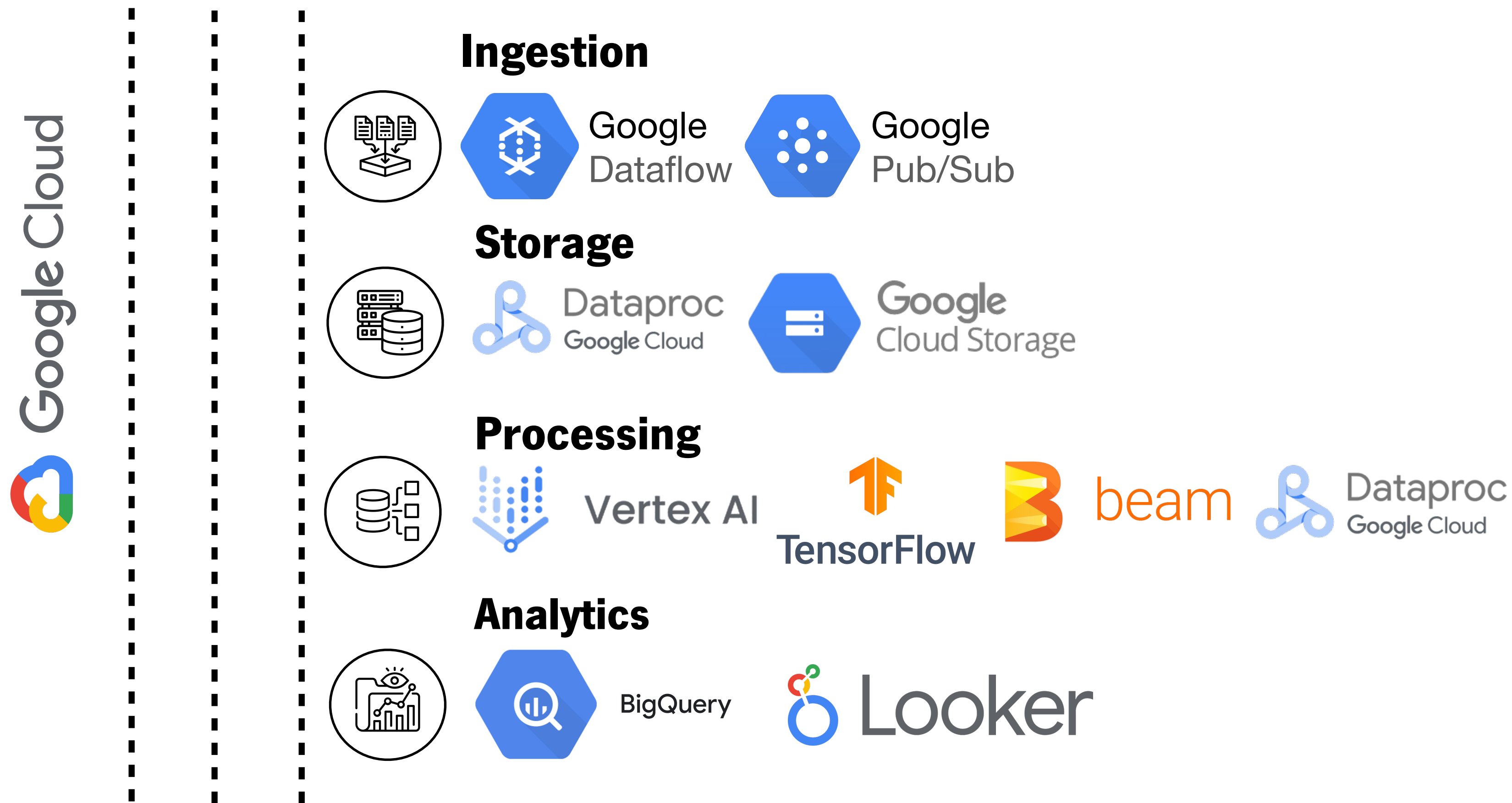
<https://netflixtechblog.com/helping-vfx-studios-pave-a-path-to-the-cloud-d308ee43c079>

<https://netflixtechblog.com/supporting-diverse-ml-systems-at-netflix-2d2e6b6d205d>

Big Data Analysis - Pipelines



❖ Use Case 4 - **Google**'s Web Page Ranking and Ads Targeting



<https://cloud.google.com/blog/topics/developers-practitioners/what-data-pipeline-architecture-should-i-use>
<https://static.googleusercontent.com/media/research.google.com/en//pubs/archive/44686.pdf>

Current Challenges

- ❖ How can one **efficiently process** terabytes of data while leveraging the **scale** and **dependability** offered by Cloud Infrastructures (i.e., Cloud Computing and High-Performance Computing)?
- ❖ How can one **design and optimize** large-scale data pipelines to ensure **efficient, scalable** and **fault-tolerant** data processing?
- ❖ What are the **best strategies** for **training** and **deploying machine learning** models in **distributed environments** while ensuring scalability and reliability?

Objectives

- ❖ Understand the challenges of analyzing large volumes of data.
- ❖ Solve the challenges of analyzing large volumes of data.
- ❖ Analyze case studies of big data analysis.
- ❖ Learn tools and best practices for more effective, scalable, robust, and reproducible data analysis.
- ❖ Design effective solutions for analyzing large volumes of data.
- ❖ Implement effective solutions for analyzing large volumes of data.

Roadmap

Theoretical

❖ Phase 1: Fundamentals of Big Data and Distributed Computing

- **Lesson 1** - Introduction to Big Data
- **Lesson 2** - Distributed Computing Paradigms
- **Lesson 3** - Processing Models

❖ Phase 2: Apache Spark and Performance Optimization

- **Lesson 4** - Apache Spark Fundamentals
- **Lesson 5** - Performance Optimization in Spark
- **Lesson 6** - Building Data Pipelines in Spark

❖ Phase 3: Distributed Machine Learning

- **Lesson 8** - Introduction to Distributed Machine Learning
- **Lesson 9** - PyTorch vs TensorFlow
- **Lesson 10** - Distributed Training with PyTorch

Roadmap

Practical

❖ Phase 1: Environment Setup & Data Manipulation

- **Lesson 1** - Setting up the Environment
- **Lesson 2** - Configuring a Spark Cluster and First Data Manipulation Scripts

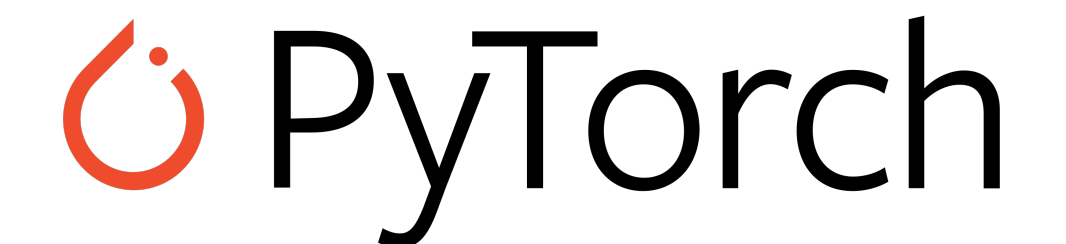
❖ Phase 2: Distributed Data Processing

- **Lesson 3** - Parallel Processing in Spark, RDDs and DataFrames in Spark
- **Lesson 4** - Optimizing Queries with Spark SQL
- **Lesson 5** - Building and Automating Data Pipelines

❖ Phase 3: Machine Learning in Distributed Systems

- **Lesson 6** - Distributed Machine Learning with Spark ML
- **Lesson 8** - Basic Models in PyTorch and TensorFlow
- **Lesson 9** - Training Distributed Models with PyTorch
- **Lesson 10** - Building a Full AI Pipeline with Spark & PyTorch
- **Lesson 11** - Benchmarking and Evaluation of Optimization

Key Technologies



Assessment

- ❖ 2 Projects (50%) – minimum grade: 10 values - **3 elements**
 - Project #1: 19/03/2026 (23:59)
 - Project #2: 07/05/2026 (23:59)
- ❖ Written exam (50%) – minimum grade: 8 values
 - 27/05/2026

Main References

- ❖ Big Data: Principles and best practices of scalable real-time data systems - Nathan Marz
- ❖ Cloud Computing: Concepts, Technology & Architecture - Thomas Erl
- ❖ Designing Data-Intensive Applications: The Big Ideas Behind Reliable, Scalable, and Maintainable Systems. Martin Kleppmann. 2017. O' Reilly Media, Inc
- ❖ High-Performance Python. Micha Gorelick, Ian Ozsvald. 2020. O' Reilly Media, Inc
- ❖ Data Science with Python and Dask. Jesse C. Daniel. 2019. Manning
- ❖ Spark: The Definitive Guide. Bill Chambers, Matei Zaharia. 2018. O' Reilly Media, Inc

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Questions?